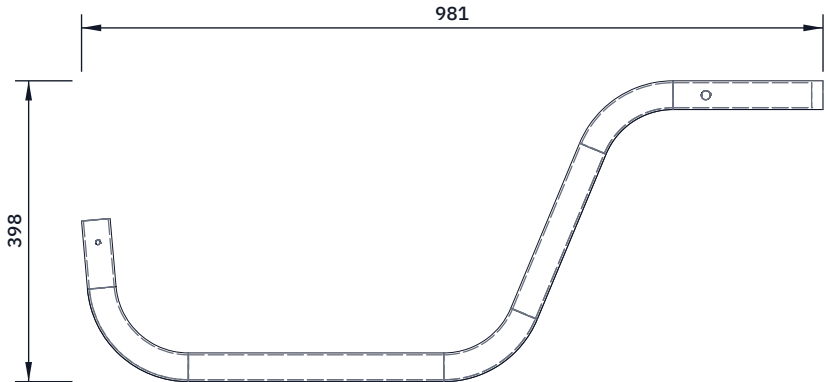


REV	DESCRIPTION	DATE	BY
P1	Making sketch for the prototype build plan	2026-10-01	AC



TOP VIEW

Scale 1:10; looking down (along -Z)



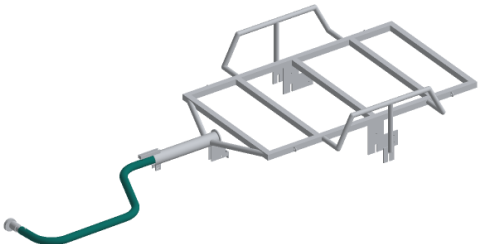
FRONT VIEW

Scale 1:10; looking at the front (along +Y)



RIGHT VIEW

Scale 1:10; looking at the right side (along -X)



WHERE IT GOES

This part in colour, its neighbours in grey

HOW TO MAKE IT AND HOW IT FITS

Cut 1,320 mm of 38 x 2.5 S355 tube (trim to 1,305 after bending).

Three bends, 115 mm centre-line radius, in a tube bender with a 38 mm die. From the hitch end: 90 mm straight, bend 85 degrees left; 338 mm straight, bend 67 degrees right; 237 mm straight, bend 67 degrees left; 198 mm straight.

The bends lie almost flat: the straight rear part sits 16 mm higher than the hitch end. Check on a table with a packer.

The rear 198 mm must be straight and round to slide in the bushings: no dents, file off any weld spatter.

Rear end: weld in a 33 mm plug, 15 mm long, tapped M10.

Overload pin hole: 12.2 mm, straight down through both walls, 155 mm from the rear end, square to the last bend's plane.

Hitch end: 6.2 mm locking pin hole, 30 mm from the end.

Check: the rear end slides through a 38.2 mm ring gauge.

CargoMule drawbar: making sketch

DRAWING NO. CGM-DWG-105			REV P1	SHEET 1 of 1	SIZE ANSI B
SCALE 1:10	UNITS mm	PROJECTION 	DATE 2026-10-01		DRAWN Amish Chadha
MATERIAL / NOTES S355 steel tube 38 x 2.5 mm					
Licensed CERN-OHL-S-2.0 · Design Molecule Lab · designmolecule.com				Generated from cad/src; do not edit by hand	